

Mach-Zender Interferometer design proposal

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Abstract: This work presents a design of a Mach-Zender Interferometer based in silicon photonics, from the simulation of the individual components of the system up to the final layout for fabrication. Experimental analysis of the fabricated sample will be performed later

1. Introduction

Mach-Zehnder interferometers (MZIs) are fundamental building blocks in silicon photonics, enabling precise control of light through interference. An MZI typically consists of two optical splitters connected by two waveguide arms. Light entering the device is divided between the arms, where a relative phase difference is introduced, and then recombined at the output. The resulting interference determines the transmitted optical power.

In silicon photonics, the phase difference can be controlled using mechanisms such as thermo-optic heating, carrier injection, or carrier depletion, allowing MZIs to function as optical modulators, switches, filters, and sensors. Their compatibility with standard silicon fabrication processes, compact footprint, and ability to integrate with electronic circuits make MZIs particularly attractive for applications in optical communications and photonic computing.

2. Theory

We can consider the electric field at the beginning each of the branches of an MZI as $E_{i1} = E_{i2} = E_i/\sqrt{2} e^{i(\omega t - \beta z)}$, where E_i is the initial field amplitude, and $\beta = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}$. At the joining of the two paths, the field will be $E_o = E_{o1} + E_{o2} = E_i e^{i\omega t} / \sqrt{2} \cdot (e^{-i\beta L_1} + e^{-i\beta L_2})$. The intensity will then be $I_o = |E_o|^2 = E_i^2/2 \cdot (1 + \cos(\beta \cdot \Delta L))$, where $\Delta L = L_2 - L_1$. This means that, for a certain ΔL , the output power oscillates with the wavelength between full transmission and 0 transmission following a cosine function, where we define the free spectral range (FSR) as the distance between consecutive transmission minima/maxima, which can be calculated as $FSR = \frac{\lambda^2}{\Delta L \cdot n_g(\lambda)}$, where $n_g = n_{\text{eff}}(\lambda) - \lambda \cdot \frac{dn_{\text{eff}}(\lambda)}{d\lambda}$ is the group index.

3. Modeling and simulation

3.1. Waveguide

The waveguides used in our design consist of silicon strip waveguides with a width of 500nm and a height of 220nm surrounded by a SiO₂ cladding. These parameters for the waveguide only allow one TE and one TM modes to propagate. Figure 1 shows both modes. The effective and group indices for both modes as a function of the wavelength are shown in Figures 2 and 3. A compact expression for n_{eff} can be obtained for each mode:

$$n_{\text{eff}} = 2.45 - 1.13 * (\lambda - 1.55) - 0.04 * (\lambda - 1.55)^2 \text{ for the TE mode}$$

$$n_{\text{eff}} = 1.77 - 1.26 * (\lambda - 1.55) + 1.91 * (\lambda - 1.55)^2 \text{ for the TM mode}$$

3.2. Layout

An image of the layout can be seen in Figure 4. There are 2 simple circuits without interferometers used to obtain the insertion losses (InsertTestE and InsertTestM for TE and TM respectively). 4 MZI have been designed, 3 TE ones (MZI1, MZI2, MZI3) and a TM one (MZIM). Apart from this, two splitters (SplitterE and Splitter M for TE and TM respectively) have been designed by

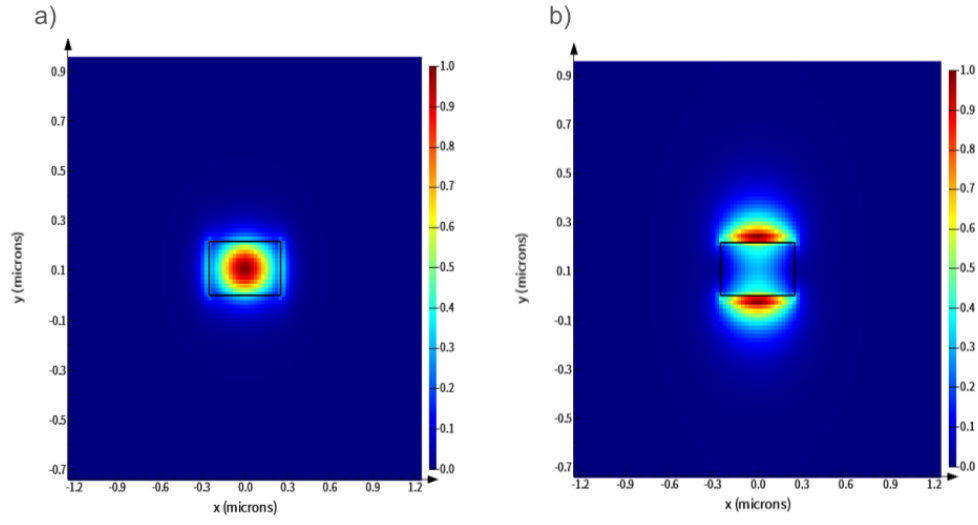


Fig. 1. Mode profile for: a) TE, and b) TM modes.

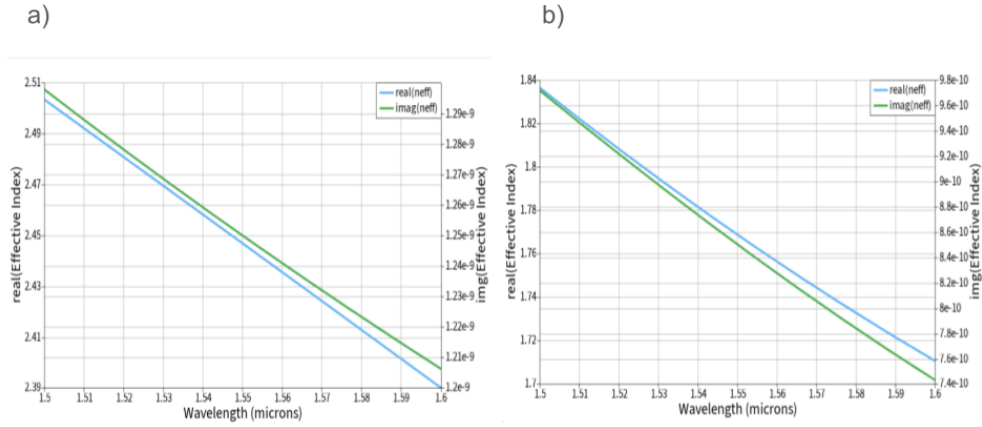


Fig. 2. Effective index for: a) TE, and b) TM modes.

43 using a very short ΔL , leading to 75/25 and 80/20 splitting approximately. Table 1 shows the
 44 path length of each circuit. Figure 5 shows the transmission for MZI3. From the FSR, n_g can be
 45 obtained from the equation in section 2.

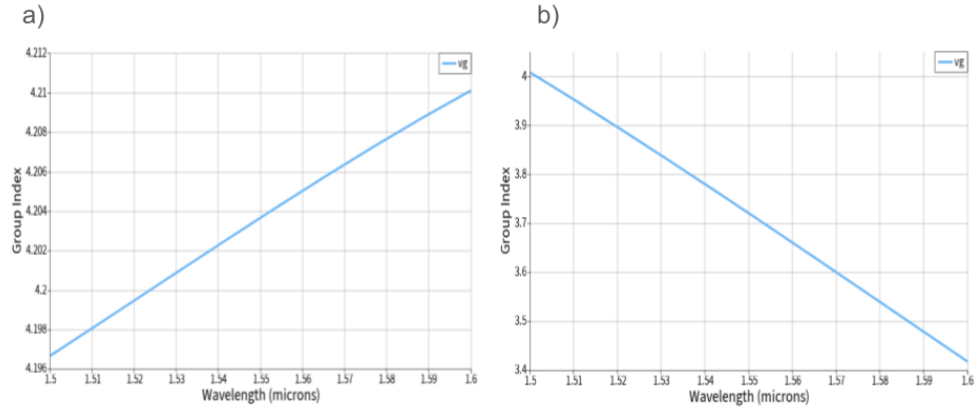


Fig. 3. Group index for: a) TE, and b) TM modes.

Table 1. Path difference ΔL for each circuit.

Label	$\Delta L(\mu\text{m})$
MZI1	70
MZI2	160
MZI3	30
MZIM	50
SplitterE	0.15
SplitterM	0.15

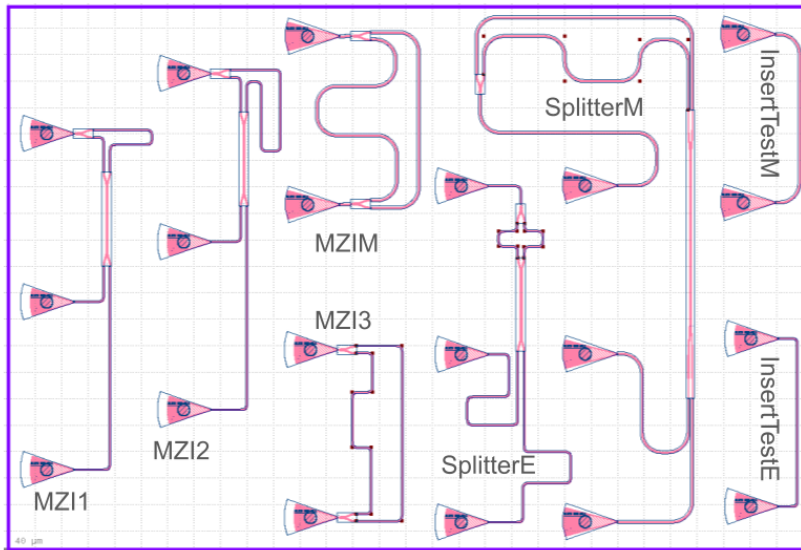


Fig. 4. Layout for fabrication

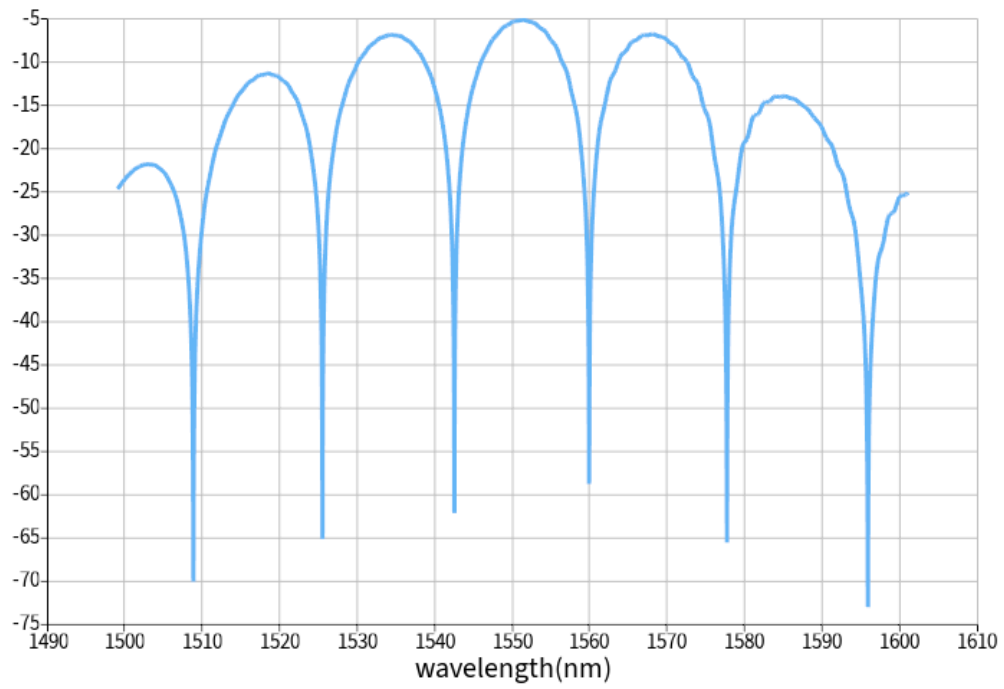


Fig. 5. Transfer function of MZI3